CS 4248
Introduction to Large Language Models

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Materials

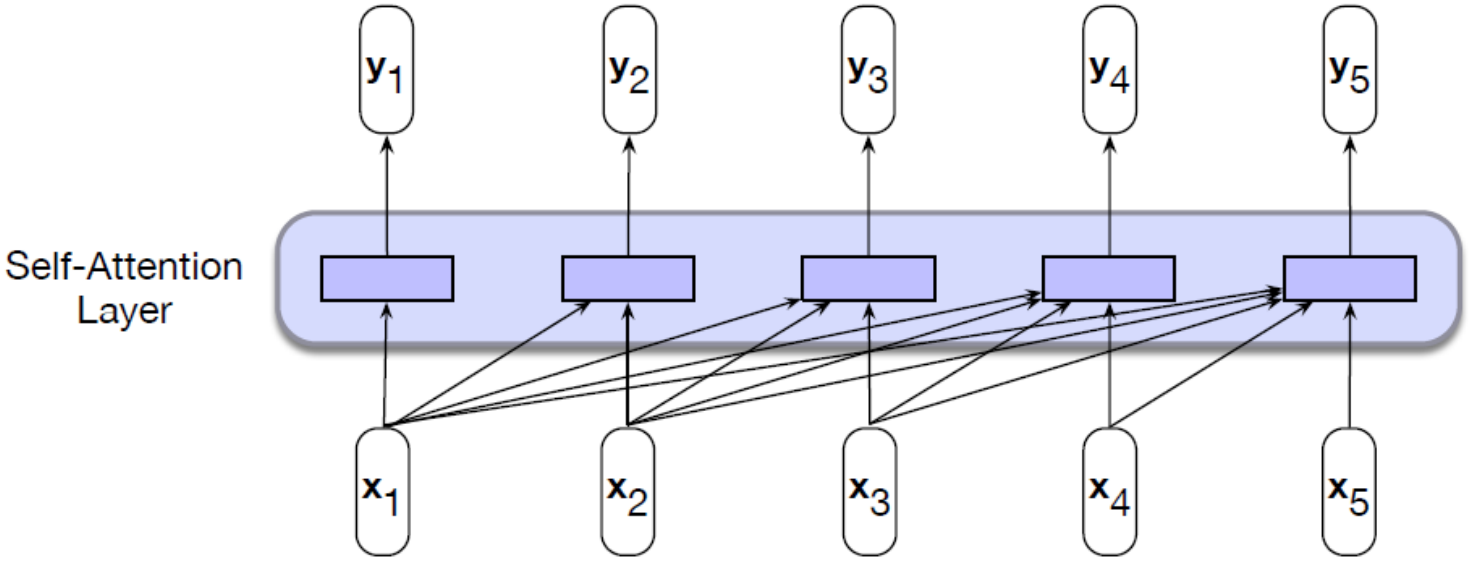
- Chapter 9 of Jurafsky and Martin
- Masked language models (encoder-only transformers)

Masked Language Models

Bidirectional Transformer Encoder

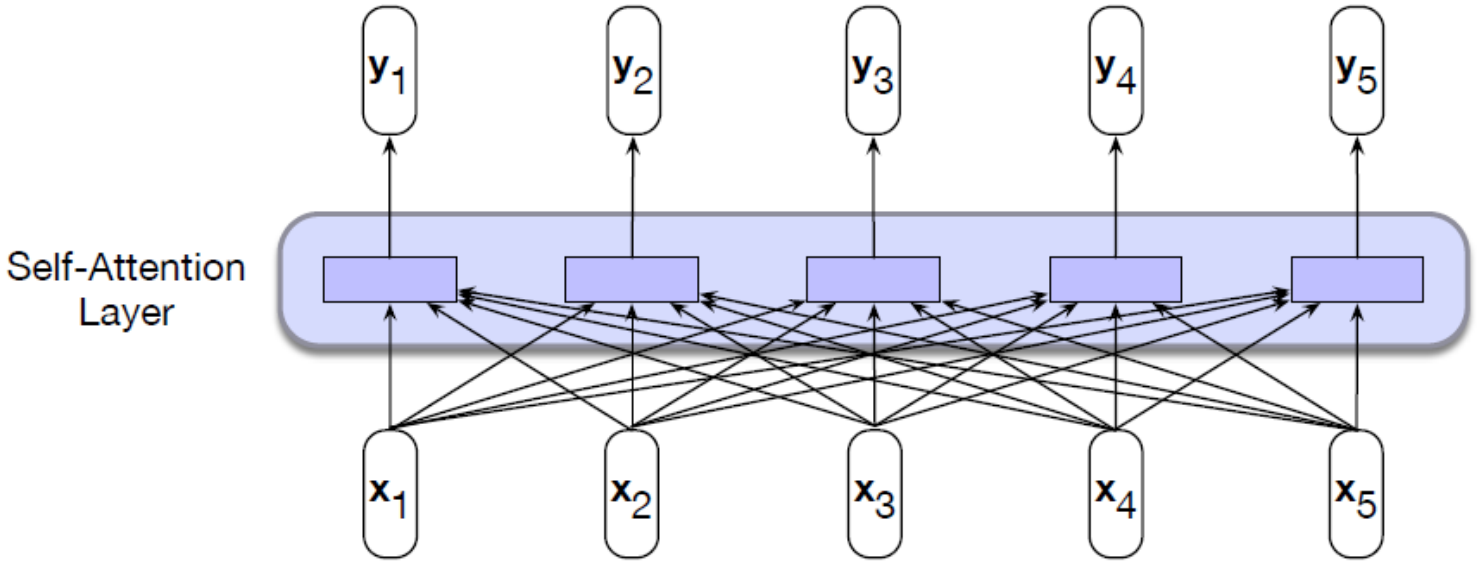
- Consider both left and right context (bidirectional)
- BERT (Bidirectional Encoder Representations from Transformers)

Self-Attention Networks



A causal (left-to-right) self-attention layer

Bidirectional Transformer Encoder



QK^T Matrix

N

q1·k1	−∞	−∞	−∞
q2·k1	q2·k2	−∞	−∞
q3·k1	q3·k2	q3·k3	−∞
q4·k1	q4·k2	q4·k3	q4·k4

N

Decoder-only transformer
(masking)

N

q1·k1	q1·k2	q1·k3	q1·k4
q2·k1	q2·k2	q2·k3	q2·k4
q3·k1	q3·k2	q3·k3	q3·k4
q4·k1	q4·k2	q4·k3	q4·k4

N

Encoder-only transformer
(no masking)

Model Architecture of BERT

- BERT: encoder-only transformer
- Input: tokens
- V: vocabulary
- N: Input context window size (number of tokens)
- L: number of transformer blocks
- d: input vector dimension
- A: number of self-attention heads
- $BERT_{BASE}$: |V| = 30,000, N = 512, L = 12, d = 768, A = 12 (110M parameters)

Bidirectional Transformer Encoder

- Transfer learning
- Two steps: pretraining then fine-tuning
- Pretraining
  - Train the transformer model on a massive amount of unlabeled text for masked language model and next sentence prediction to get contextualized representation

Bidirectional Transformer Encoder

- Fine-tuning
  - Add one additional output layer for downstream (sentence-level and token-level) NLP tasks
  - Initialize the transformer model with the pretrained parameters
  - All parameters are fine-tuned using a smaller amount of labeled training data from the downstream NLP task

Training Bidirectional Encoders

Pretraining BERT

- Predict a fill-in-the-blank task (cloze task)
- Example:
  - “Please turn your _____ in on Friday.”
- Missing one or more tokens in the input sequence
- Denoising: Other generalized perturbations to input text include deletion, insertion, reordering, etc.

Pretraining BERT

- Masked language model
  - Randomly choose 15% of all input tokens to be manipulated
  - Of these chosen tokens (e.g., delicious)
    - 80%: replaced with [MASK] (lunch was delicious → lunch was [MASK])
    - 10%: replaced with another random token (lunch was delicious → lunch was gasp)
    - 10%: unchanged (e.g., lunch was delicious → lunch was delicious)
  - Model to predict the correct token for these manipulated tokens with cross-entropy loss

BERT Input Representation

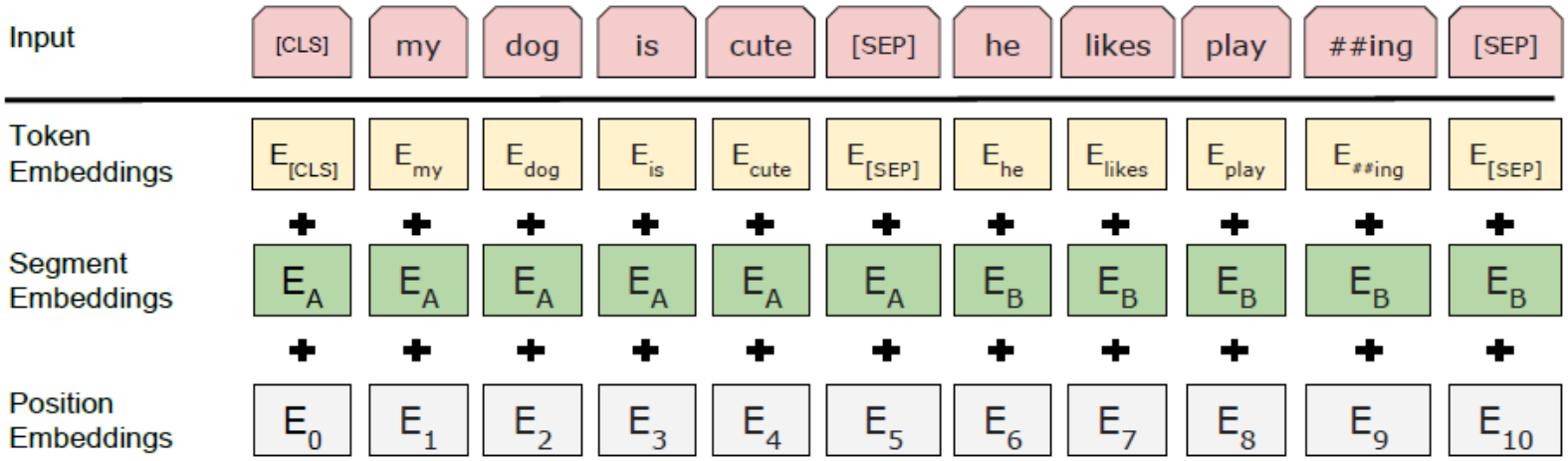
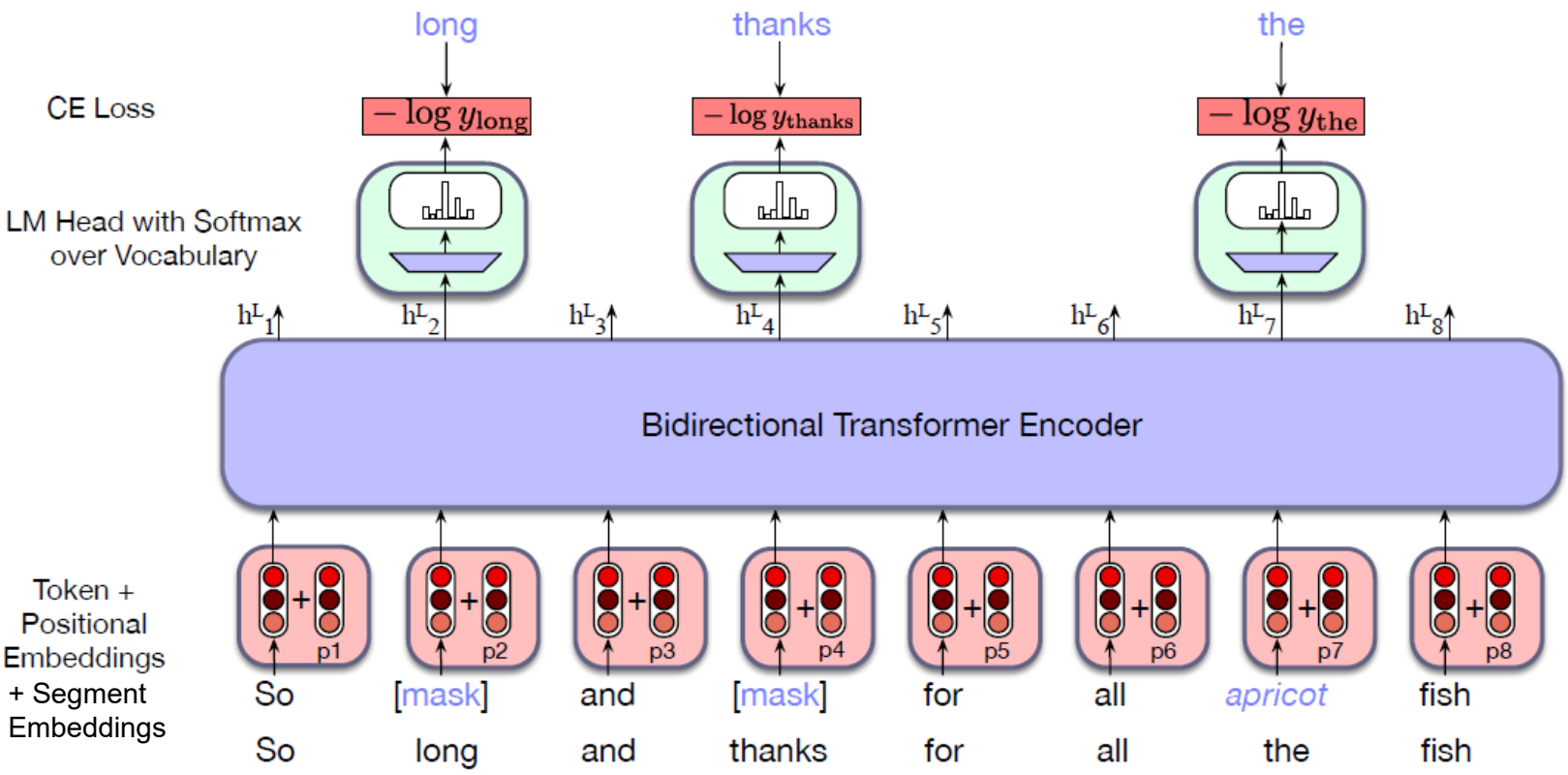


Diagram from the BERT paper, NAACL 2019

Pretraining BERT



Pretraining BERT

$$\boldsymbol{u}_i = \boldsymbol{h}_i^L \boldsymbol{E}^T$$

$$\mathbf{y}_i = \mathrm{softmax}(\boldsymbol{u}_i)$$

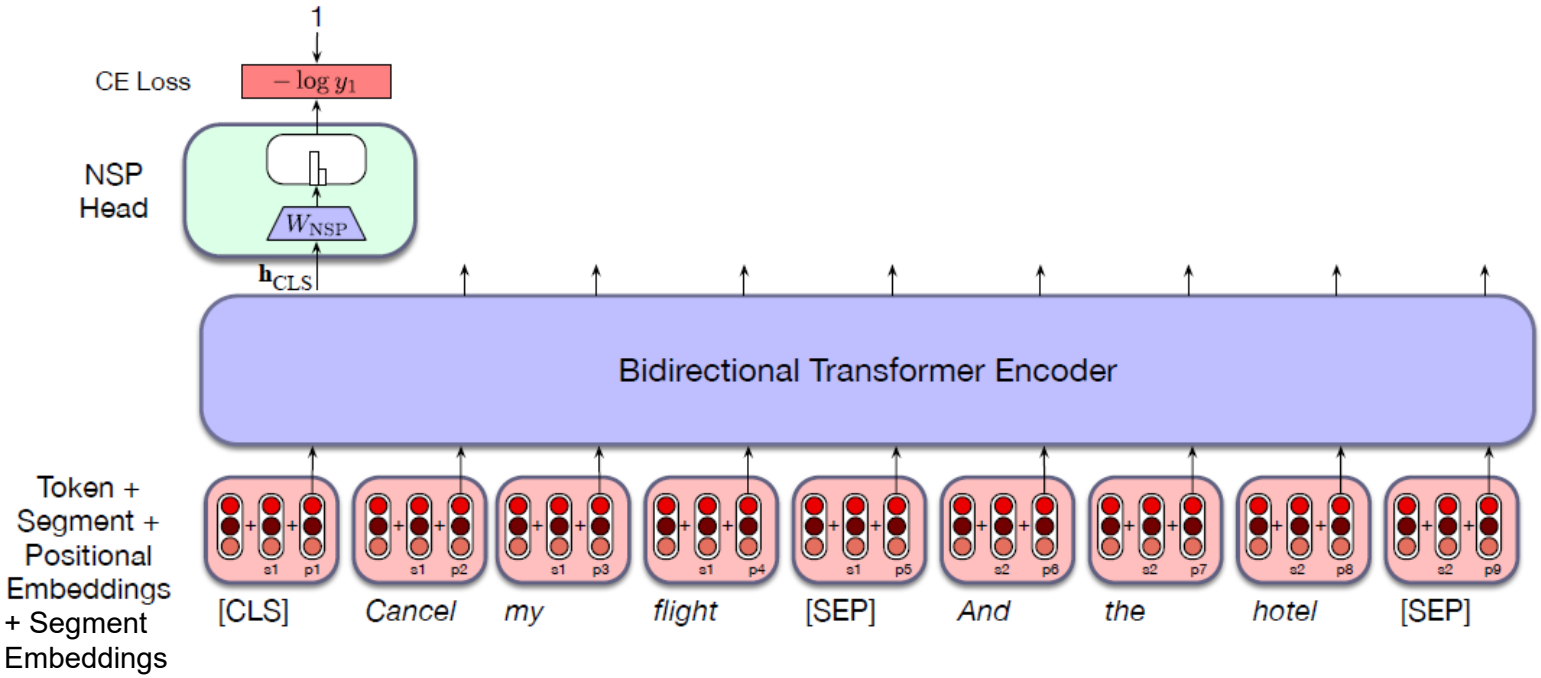
$$L_{\mathrm{MLM}} = -\frac{1}{|M|}\sum_{i \in M} \log P(x_i | \boldsymbol{h}_i^L)$$

M: the set of masked tokens
x_i: the correct token at position i

Pretraining BERT

- Next sentence prediction
  - Choose sentences A and B for each pretraining example
  - IsNext: B is the actual next sentence that follows A
  - NotNext: B is a random sentence from the corpus
  - Two-class prediction based on [CLS]
- Cross-entropy loss

Pretraining BERT



$$\boldsymbol{y} = \text{softmax}\left(\boldsymbol{h}_{\text{CLS}}^{L}\boldsymbol{W}_{\text{NSP}}\right) \qquad \boldsymbol{W}_{\text{NSP}} \in \mathbb{R}^{d\times 2}$$

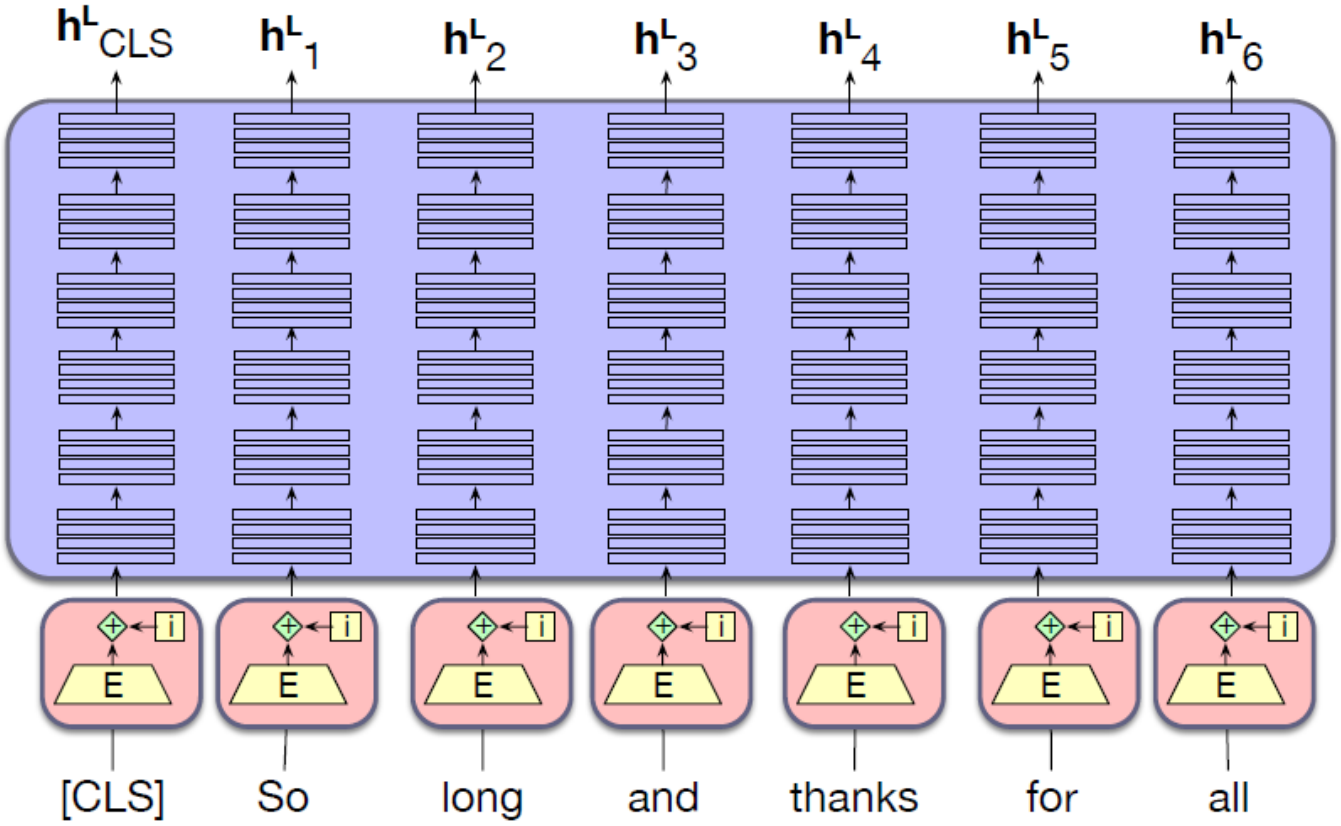
Pretraining Corpora for BERT

- BooksCorpus (800M words)
- English Wikipedia (2.5B words)
- Total size = 3.3B words

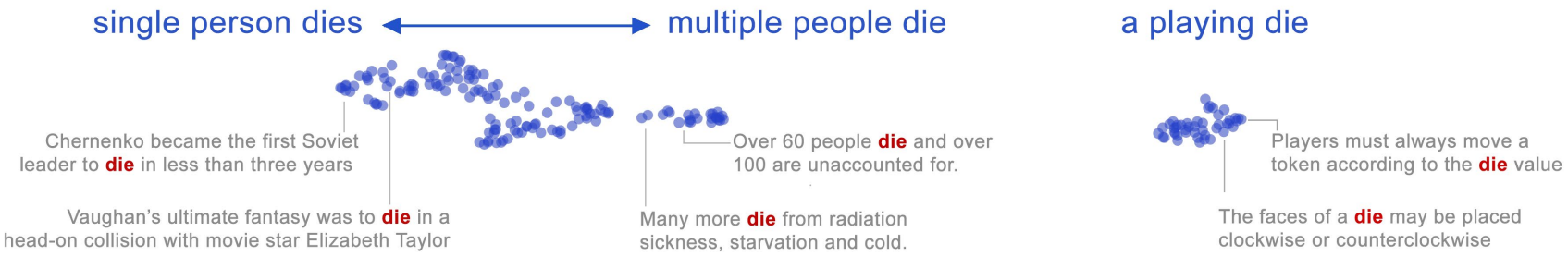
Contextual Embeddings

- Given pretrained BERT and a new input sentence of tokens $x_1, \ldots, x_N$, the output vector $\boldsymbol{h}_i^L$ from the final layer of BERT is a contextual embedding or representation of x_i in the context of the input sentence

Contextual Embeddings



Contextual Embeddings and Word Sense



Benefits of Masked Word Prediction

- She bought many fruits like apples, ___, pears, and bananas at the supermarket. (lexical semantics)
- David and ____ wife were divorced. (coreference)
- The restaurant was _____ – The food was delicious, the service was top-notch, and the price was low. (sentiment)

Fine-Tuning

Fine-Tuning BERT

- Sentence-level task (sequence classification and sentence-pair classification)
  - Final output vector for the [CLS] token
  - Sentiment classification
  - Natural language inference (NLI)
- Token-level task (sequence labeling or sequence tagging)
  - Final output vector for the ith input token
  - Named entity recognition

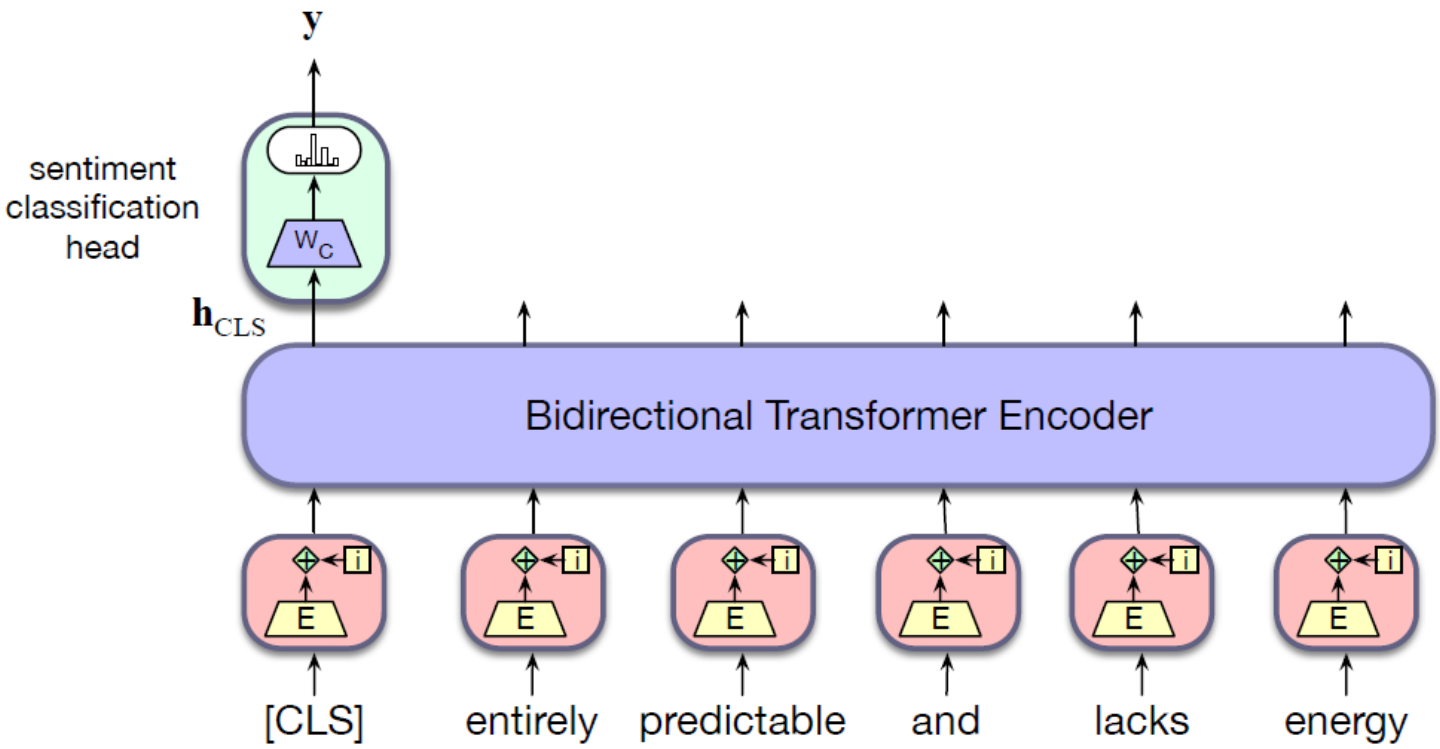
Fine-Tuning BERT

- Update the weights of the pretrained language model and the weights of the classifier head
- Require supervised training data of the NLP application

Sequence Classification

- Sentiment classification
- Spam detection
- Topic classification

Sequence Classification



$$\boldsymbol{y} = \text{softmax}\left(\boldsymbol{h}_{\text{CLS}}^{L}\boldsymbol{W}_C\right) \qquad \boldsymbol{W}_C \in \mathbb{R}^{d \times N_c}$$

Sentence-Pair Classification

- Sentence-pair classification tasks
  - Paraphrase detection (paraphrase or not)
  - Natural language inference (entailment, contradiction, no relation)
  - Discourse coherence (how coherent is sentence B as a follow-on to sentence A)
- Degenerate text – ∅ pair in sequence classification or sequence labeling

Natural Language Inference

- Determine the relationship between the first sentence (premise) and the second sentence (hypothesis)
- Entailment
  - I am confused.
  - Not all of it is very clear to me.
- Contradiction
  - Tourist information offices can be very helpful.
  - Tourist information offices are never of any help.
- No relation
  - John travelled back to his hometown.
  - Mary was tired.

Sequence Labeling

- Name entity
  - Proper name

Type	Tag	Sample Categories	Example sentences
People	PER	people, characters	**Turing** is a giant of computer science.
Organization	ORG	companies, sports teams	The **IPCC** warned about the cyclone.
Location	LOC	regions, mountains, seas	**Mt. Sanitas** is in **Sunshine Canyon**.
Geo-Political Entity	GPE	countries, states	**Palo Alto** is raising the fees for parking.

- Named entity recognition (NER)
  - Find spans of text that constitute proper names and tag the type of the entity

Sequence Labeling

- Challenge: Name type ambiguity

[$_{PER}$ Washington] was born into slavery on the farm of James Burroughs.
[$_{ORG}$ Washington] went up 2 games to 1 in the four-game series.
Blair arrived in [$_{LOC}$ Washington] for what may well be his last state visit.
In June, [$_{GPE}$ Washington] passed a primary seatbelt law.

Sequence Labeling

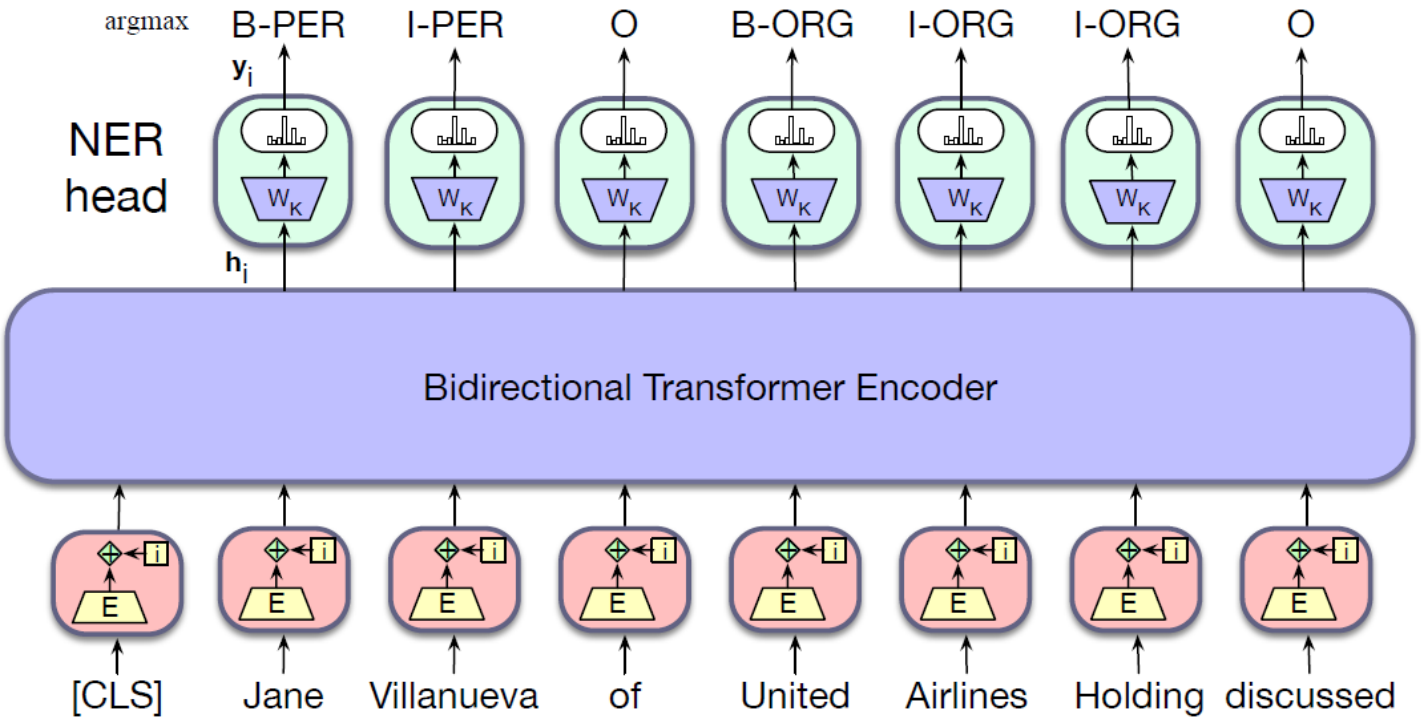
[$_{PER}$ Jane Villanueva] of [$_{ORG}$ United Airlines Holding] discussed the [$_{LOC}$ Chicago] route .

- Tagging schemes
  - IO tagging
  - BIO tagging
  - BIOES tagging
- Perform NER by assigning a tag to each word

Sequence Labeling

Words	IO Label	BIO Label	BIOES Label
Jane	I-PER	B-PER	B-PER
Villanueva	I-PER	I-PER	E-PER
of	O	O	O
United	I-ORG	B-ORG	B-ORG
Airlines	I-ORG	I-ORG	I-ORG
Holding	I-ORG	I-ORG	E-ORG
discussed	O	O	O
the	O	O	O
Chicago	I-LOC	B-LOC	S-LOC
route	O	O	O
.	O	O	O

Sequence Labeling



$$\boldsymbol{y}_i = \text{softmax}(\boldsymbol{h}_i^L \boldsymbol{W}_K) \qquad \boldsymbol{W}_K \in \mathbb{R}^{d \times N_c}$$

$$\boldsymbol{t}_i = \text{argmax}_k(\boldsymbol{y}_i)$$

Extractive QA

- Sentence A and B: Question-passage pairs in question answering
- Answer span extraction
- Input: a passage and a question
  - Passage: "… Reaching 29,029 feet at its summit, Mt. Everest is the highest mountain in the world …"
  - Question: "How tall is Mt. Everest?"
  - Extract a contiguous segment of text (span) in the passage as the answer
  - Answer: 29,029 feet

Span Labeling

- Input:
  - Question q of n tokens: $q_1, \ldots q_n$
  - Passage p of m tokens: $p_1, \ldots p_m$
- Compute $P(a|q,p)$ for each possible span a, where a starts at position a_s and ends at position a_e
- $P(a|q,p) = P_s(a_s|q,p)P_e(a_e|q,p)$

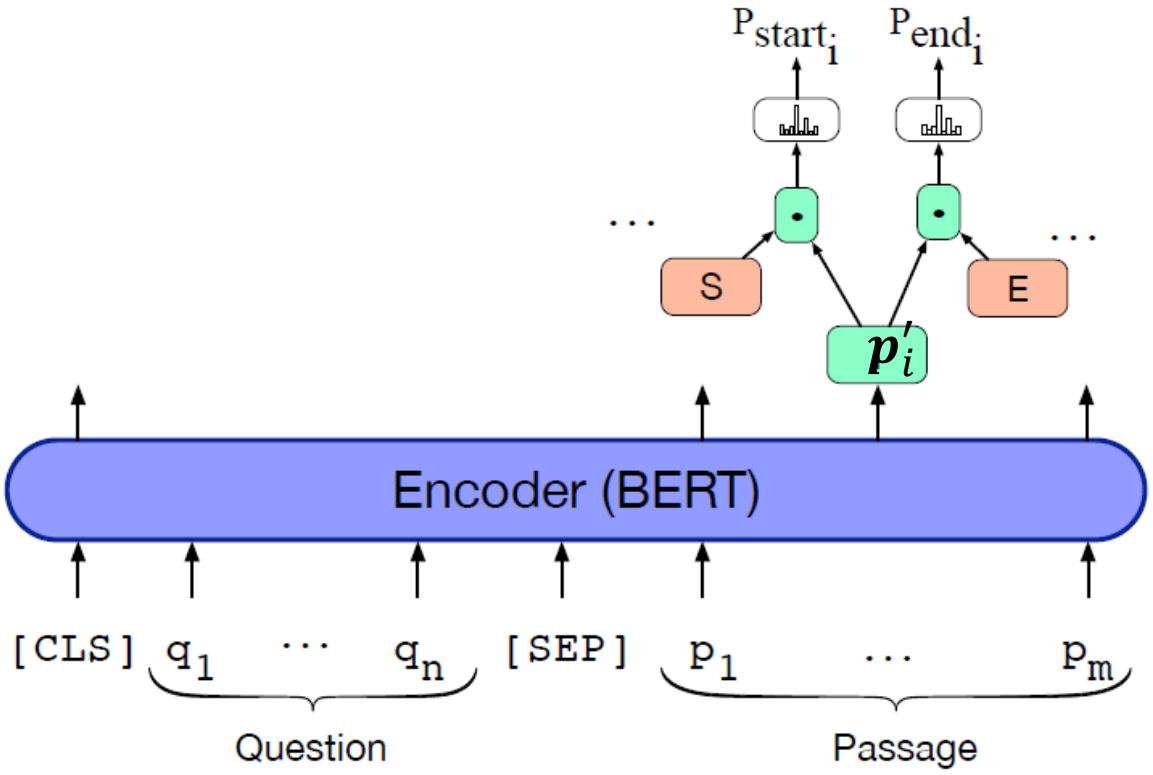
Span Labeling

- Add a linear layer on top of BERT with two new vectors $\boldsymbol{s}$ and $\boldsymbol{e}$.
- $\boldsymbol{p}_i'$: BERT output vector of token p_i in passage

$$P_s(i) = \frac{\exp(\boldsymbol{s} \cdot \boldsymbol{p}_i')}{\sum_{j=1}^{m} \exp(\boldsymbol{s} \cdot \boldsymbol{p}_j')}$$

$$P_e(i) = \frac{\exp(\boldsymbol{e} \cdot \boldsymbol{p}_i')}{\sum_{j=1}^{m} \exp(\boldsymbol{e} \cdot \boldsymbol{p}_j')}$$

Span Labeling



Span Labeling

- Loss function $L = -\log P_s(a_s^*) - \log P_e(a_e^*)$ where a_s^* and a_e^* are the start and end position of the correct answer span
- During inference, score of a candidate span from position i to j is $\boldsymbol{s} \cdot \boldsymbol{p}_i' + \boldsymbol{e} \cdot \boldsymbol{p}_j'$
- Highest scoring span where $i \leq j$ is the predicted answer span

Span Labeling

- BERT with a linear layer added on top is a standard baseline model for extractive QA.
- Extensions:
  - Answer is not present in the passage (e.g., SQuAD 2.0): Returns [CLS] token as the “answer”.
  - Long passage: Break up a long passage into overlapping shorter passages and perform QA with the question and each shorter passage. No answer is missed as long as the answer span is shorter than the length of the overlap.